CHARACTERIZATIONS OF SCHEMES OF DIMENSION ZERO

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file properties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the cycle of implications through the criteria on affine open coverings, and the two closing deductions: connected components are points via separation by an affine open, and quasi-compact opens are affine by induction using the three-piece decomposition.

Editorial note. The label lemma-dimension-zero names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source properties.tex –label lemma-dimension-zero –start 1519 –end 1587 –local-id 28031 returned RESULT=OK (L28031 free, occupancy FREE, public interval clear). Candidate taken from sources/stacks-properties/candidates-halfA.json (third exclusive pool assigned by the Lead on 2026-10-13).

Editorial note. The statement carries a native bibliography pointer environment naming the source of the argument; that environment is part of the delivered statement and is covered by the preamble span 8-13 of the chapter, which this card includes.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Structure of the spectrum of a ring with only minimal primes. Algebra, the lemma describing the spectrum of a ring whose primes are all minimal as a profinite space with the properties used freely in the proof. Supplies the properties of an affine open of a scheme of dimension zero, in particular its profiniteness and Hausdorffness.
- Criteria for quasi-separatedness and separatedness in terms of intersections of affine opens. Schemes, the lemmas characterizing quasi-separated and separated schemes through the intersections of the members of an affine open covering. Supplies the two characterizations applied in the first two implications.
- Quasi-compact subsets of a Hausdorff space are closed. Topology, the lemma that a quasi-compact subset of a Hausdorff space is closed. Supplies the implication from Hausdorffness to the closedness of affine opens.

Source reference. Original lemma 04MG: sources/stacks-chow/tags.txt, source lines 6042–6042. Prerequisite pointer invoked at proof line(s) 1540; it names a standard

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fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 01KO: sources/stacks-chow/tags.txt, source lines 2057–2057. Prerequisite pointer invoked at proof line(s) 1546; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 01KP: sources/stacks-chow/tags.txt, source lines 2058–2058. Prerequisite pointer invoked at proof line(s) 1556; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 08YB: sources/stacks-chow/tags.txt, source lines 11334–11334. Prerequisite pointer invoked at proof line(s) 1571; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let X be a scheme of dimension zero. The following are equivalent*

(1) *X is quasi-separated,*
(2) *X is separated,*
(3) *X is Hausdorff,*
(4) *every affine open is closed.*

In this case the connected components of X are points and every quasi-compact open of X is affine. In particular, if X is quasi-compact, then X is affine.

Proof. As the dimension of X is zero, we see that for any affine open $U \subset X$ the space U is profinite and satisfies a bunch of other properties which we will use freely below, see Algebra, Lemma 04MG. We choose an affine open covering $X = \bigcup U_i$.

If (4) holds, then $U_i \cap U_j$ is a closed subset of U_i, hence quasi-compact, hence X is quasi-separated, by Schemes, Lemma 01KO, hence (1) holds.

If (1) holds, then $U_i \cap U_j$ is a quasi-compact open of U_i hence closed in U_i. Then $U_i \cap U_j \to U_i$ is an open immersion whose image is closed, hence it is a closed immersion. In particular $U_i \cap U_j$ is affine and $\mathcal{O}(U_i) \to \mathcal{O}_X(U_i \cap U_j)$ is surjective. Thus X is separated by Schemes, Lemma 01KP, hence (2) holds.

Assume (2) and let $x, y \in X$. Say $x \in U_i$. If $y \in U_i$ too, then we can find disjoint open neighbourhoods of x and y because U_i is Hausdorff. Say $y \notin U_i$ and $y \in U_j$. Then $y \notin U_i \cap U_j$ which is an affine open of U_j and hence closed in U_j. Thus we can find an open neighbourhood of y not meeting U_i and we conclude that X is Hausdorff, hence (3) holds.

Assume (3). Let $U \subset X$ be affine open. Then U is closed in X by Topology, Lemma 08YB. This proves (4) holds.

Assume X satisfies the equivalent conditions (1) – (4). We prove the final statements of the lemma. Say $x, y \in X$ with $x \neq y$. Since y does not specialize to x we can choose $U \subset X$ affine open with $x \in U$ and $y \notin U$. Then we see that $X = U \amalg (X \setminus U)$ is a decomposistion into open and closed subsets which shows that x and y do not belong to the same connected component of X. Next, assume $U \subset X$ is a quasi-compact open. Write $U = U_1 \cup \ldots \cup U_n$ as a union of affine opens. We will prove by induction on n that U is affine. This immediately reduces us to the case $n = 2$. In this case we have $U = (U_1 \setminus U_2) \amalg (U_1 \cap U_2) \amalg (U_2 \setminus U_1)$ and the arguments above show that each of the pieces is affine. $\square$